CATO BOLAM
CONSULTANTS

R C and D E Nicklin

Tunnel Road
Puhoi, Auckland

Stormwater Hydrology Report

Our Ref: 40320

Date: 17 October 2017

Report Prepared by:

Stan Thompson
Engineer - Hydrology

Report Reviewed by:

Myles Goodwin
Principal

Contents

1.0 Introduction **1**

2.0 Site Description and Proposal **1**

3.0 Hydrological Neutrality **1**

4.0 Discharge Once Developed **2**

5.0 Stormwater Disposal **2**

6.0 Level Spreader Dispersal Device **3**

7.0 Native Bush Protection Mitigates Developments **3**

8.0 Conclusions and Recommendations **3**

9.0 Limitations **4**

Appendix A: Site and Hydrology Plans 5

Appendix B: TP108 Rainfall Calculations 6

Appendix C: Level Spreader Device Diagrams 7

Appendix D: Stormwater Device Maintenance Plan 8

1.0 Introduction

This report is provided in support of the subdivision and resource consent application for the property at Tunnel Road, Puhoi creating seven rural lots and protecting areas of native bush.

2.0 Site Description and Proposal

The 301ha property consists of several rural titles with road frontages to Remiger Road in the north and centre, and paper Tunnel Road in the south.

There are areas of moderate sloping land in the north and strong to very strong slopes across the remainder. A dwelling and ancillary buildings are present in the north east and the land cover is a mixture of pasture and areas of bush.

Overland flow paths/gullies tend north east with drainage eventually to the Puhoi River.

The proposal is to create Lots 1 to 8 and protect areas of native bush.

Developments comprising a driveway, paving, dwelling and outbuildings are anticipated on proposed Lots 1 to 5, the existing dwellings will be contained within Lots 6 and 8, and Lot 7 will be retained as the balance rural land. It is intended to provide access to Lots 1 to 5 over the existing driveway from Remiger Road in the north. The driveway will become a right of way, as shown on the site plan (Appendix A).

The runoff from the roof areas of the new dwellings on Lots 1 to 5 will flow to water reuse rain tanks. The driveway/paving will be constructed with cross fall and the runoff allowed to sheetflow over pasture to natural drainage, or may be collected in catchpits and piped to a level spreader device.

The overflow from the roof water reuse tank and flows from the driveway and paving catchpits, where used, will be piped to a level spreader dispersal device for sheetflow treatment over pasture.

It is proposed to protect in perpetuity approximately 53ha of native bush to mitigate the effects on runoff from impermeable developments anticipated on the property.

In summary, the proposed development will result in the formation of five new rural residential sites and the protection of 53ha of bush. A subdivision site plan and aerial image of the natural hydrology are shown in attached Appendix A.

3.0 Hydrological Neutrality

Auckland Council's document "Management of Stormwater in Countryside Living (Rural and Town) Zones – A Toolbox of Methods" outlines the reasons hydrological neutrality is required for new developments. The document sets out a number of principles that govern how hydrological neutrality can be achieved:

- Stormwater disposal should as practically possible mimic natural drainage.
- Modifications to existing natural drainage patterns should be kept to a minimum.
- Stormwater should not be discharged directly into streams from piped systems.
- Impervious areas should be kept to a minimum.

- Methods should be employed to detain rainfall before dispersal into waterways.

The specific aspects of hydrological neutrality that are to be addressed include the rate of flow, the volume of flow and time of concentration. The Auckland Council requires hydrological neutrality in the 100 year storm.

4.0 Discharge Once Developed

The main pattern of stormwater runoff will not change upon development, however the stormwater runoff generated will increase due to the dwellings, outbuildings and driveway/paving to be introduced on the new lots. Conversely, protection of the native bush will reduce stormwater runoff providing positive ecological effects.

We have prepared an analysis based on the existing state of the 302ha property and the post development occupancy comprising pasture, impermeable areas on Lots 1 to 8 and protected bush. The curve numbers used in the TP108 runoff analysis were 74 for pasture and unprotected bush, 98 for roof, paving (impermeable areas), 89 for gravel and 70 for the protected native bush.

A summary of TP108 calculations follows, indicating pre and post development rainfall runoff in the annual recurrence interval (ARI) storms noted.

Table 1 - TP108 Runoff Analysis Results by Storm			
State of site	**2 Year ARI (m^3)**	**10 Year ARI (m^3)**	**100 Year ARI (m^3)**
Existing	**169,362**	**382,888**	**662,055**
Developed	**166,746**	**378,657**	**656,627**
Runoff Reduction	**-2,617**	**-4,231**	**-5,428**

The analysis shows that overall runoff from the property reduces significantly in all three storm events with the proposed impermeable surfaces and as vegetation improvement occurs due to protection of native bush vegetation.

The TP108 calculations are shown in attached worksheets (Appendix B).

5.0 Stormwater Disposal

The runoff from the dwelling roof areas on proposed Lots 1 to 5 will flow to domestic storage tanks with a minimum storage capacity of 22,500 litres and water consumption by the occupants of the dwellings will most of the time provide peak rainfall storage in the rain tanks, preventing peak rainfall from becoming runoff. This makes the proposal more conservative.

The overflow outlet from the roof water tank will be connected to a level spreader dispersal device and the discharge allowed to sheetflow over pasture to natural drainage.

The driveway/paving on Lots 1 to 5 will be constructed with crossfall and where practical the runoff will be collected in catchpits and piped to a level spreader dispersal device or may optionally be allowed to sheetflow over pasture to natural drainage.

A maintenance plan for stormwater devices is shown in Appendix C.

6.0 Level Spreader Dispersal Device

The overflow from the roof water reuse tank and flows from the driveway and paving catchpits, where used, will be piped to a level spreader dispersal device for sheetflow treatment over pasture.

The dispersal device must be situated well clear of the building platform, waste water fields and steep slopes that could contribute to land instability or erosion and will be a dispersal trench or an above ground Tbar used to dissipate concentrated flows and discharge velocities to allow for even sheetflow over pasture to drainage.

The device selected will depend on the practicality for the site. The dispersal device must be installed in compliance with any geotechnical requirements and parallel to the contour to achieve an even discharge over its length to avoid scouring and erosion from concentrated flows.

Level spreader diagrams and a stormwater devices maintenance plan are attached (Appendix C).

7.0 Native Bush Protection Mitigates Developments

The proposed protection in perpetuity of 53ha of native bush mitigates up to 106,000m^2 of impermeable developments, based on the provisions of the Bush Revegetation section of the publication Countryside Living Toolbox, Auckland Region – Stormwater Management Device Design Details, April, 2010 – reference 4.7.6 Design Sizing Table C17.

Developments anticipated on the property will have area of up to 4,500m^2, comprising roof areas of 2,250m^2 and driveway/paving 2,250m^2.

8.0 Conclusions and Recommendations

The proposed subdivision results in the creation of five new rural residential sites with provision for dwellings, outbuildings, driveways and paving, and the protection of areas of native bush vegetation.

The effects on rainfall runoff from the proposed developments on the site will be mitigated by the protection of native bush areas as detailed.

Stormwater runoff from the new dwellings on Lots 1 to 5 will flow to roof water reuse tanks on each site and the overflow will be piped for sheetflow discharge over pasture via a level spreader dispersal device to natural drainage.

Runoff from the driveway/paving on Lots 1 to 5 will bypass the tank system and where practical will be collected in catchpits and piped to a level spreader dispersal device or may optionally be allowed to sheetflow over pasture to natural drainage.

The stormwater tank and other devices should be installed in compliance with any Geotechnical requirements, the manufacturers' specification, and be maintained as detailed.

Stormwater calculations shown in this report are provided to comply with Auckland Council's stormwater management rules for property development and are not intended to provide an estimate of adequacy for the domestic, firefighting or any other water supply for the property.

It has been demonstrated that hydrological neutrality can be provided for the application and we recommend that the Auckland Council adopt this report prepared in terms of the document "Management of Stormwater in Countryside Living (Rural and Town) Zones – A Toolbox of Methods" and the proposed methods of mitigating the effects of the development on stormwater discharge.

Auckland Council shall make the protection of the detailed native bush areas a condition of subdivision consent and place a consent notice on the relevant certificates of title referencing this report to satisfy the stormwater disposal requirements.

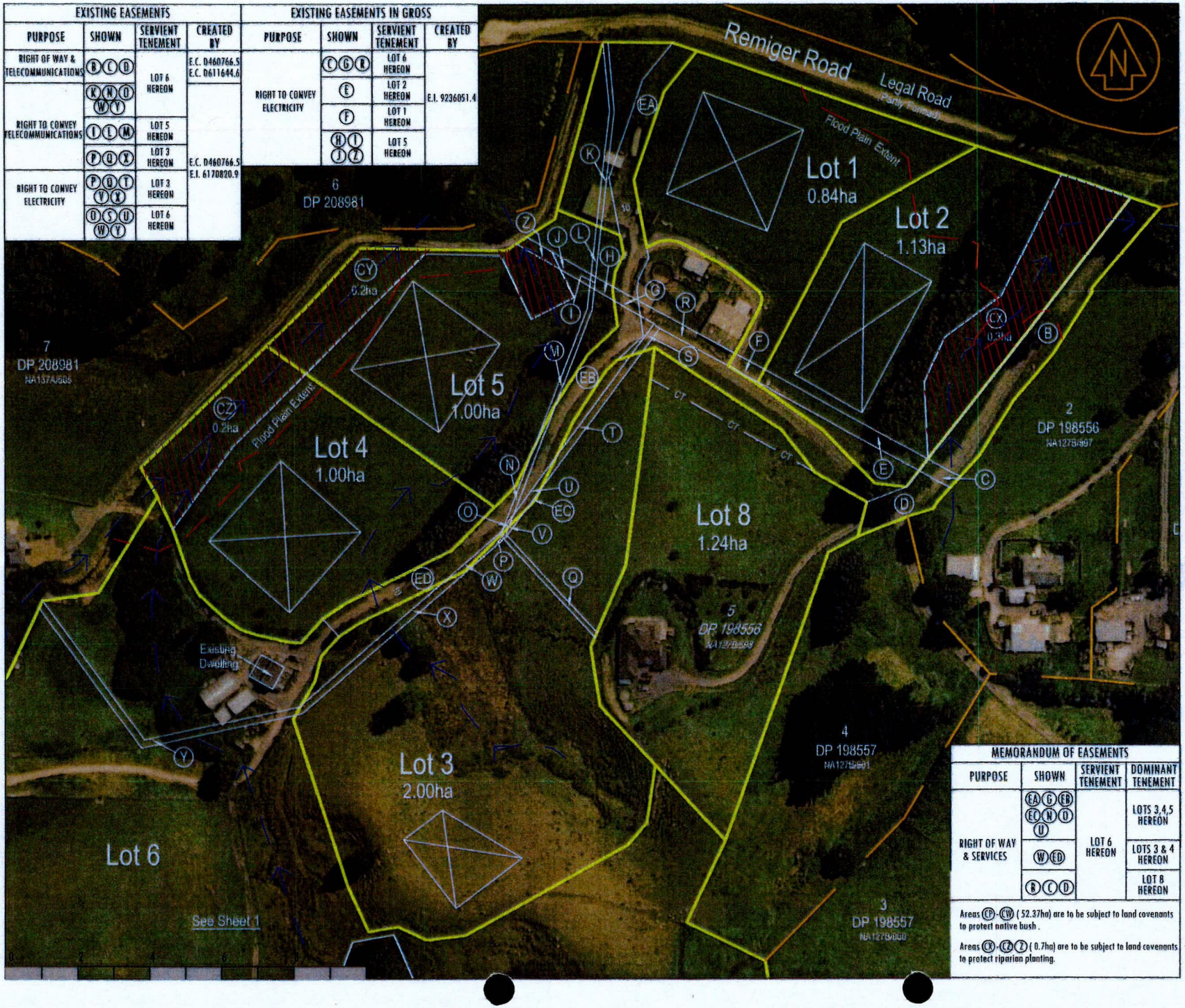
9.0 Limitations

This report has been prepared for the particular project described to us and its extent is limited to the scope of work agreed between the client and Cato Bolam Consultants Limited.

No responsibility is accepted by Cato Bolam Consultants Limited or its directors, servants, agents, staff or employees for the accuracy of information provided by third parties and/or the use of any part of this report in any other context or for any other purposes.

This report is for the use by the client only, and should not be used or relied upon by any other person or entity or for any other projects.

Appendix A: Site and Hydrology Plans



EXISTING EASEMENTS			
PURPOSE	SHOWN	SERVIENT TENEMENT	CREATED BY
RIGHT OF WAY & TELECOMMUNICATIONS	B C D	LOT 6 HEREON	E.C. D460766.5 E.C. D611644.6
RIGHT TO CONVEY TELECOMMUNICATIONS	K N O W Y	LOT 6 HEREON	E.C. D460766.5 E.I. 6170820.9
	I L M	LOT 5 HEREON	
	P Q X	LOT 3 HEREON	
RIGHT TO CONVEY ELECTRICITY	P Q T V X	LOT 3 HEREON	
	O S U W Y	LOT 6 HEREON	

EXISTING EASEMENTS IN GROSS			
PURPOSE	SHOWN	SERVIENT TENEMENT	CREATED BY
RIGHT TO CONVEY ELECTRICITY	C G R	LOT 6 HEREON	E.I. 9236051.4
	E	LOT 2 HEREON	
	F	LOT 1 HEREON	
	H I J Z	LOT 5 HEREON	

MEMORANDUM OF EASEMENTS			
PURPOSE	SHOWN	SERVIENT TENEMENT	DOMINANT TENEMENT
RIGHT OF WAY & SERVICES	EA G EB EC N O U	LOT 6 HEREON	LOTS 3,4,5 HEREON
	W ED		LOTS 3 & 4 HEREON
	B C D		LOT 8 HEREON

Areas CP-CW (52.37ha) are to be subject to land covenants to protect native bush .

Areas CX-CZ-Z (0.7ha) are to be subject to land covenants to protect riparian planting.

This drawing and design remains the property of Cato Bolam Consultants ltd and may not be reproduced without the written permission of Cato Bolam Consultants ltd.

This plan and accompanying report(s) have been prepared for the purpose of obtaining a resource consent only and for no other purpose. Use of this plan and/or the information on it for any other purpose is at the user's risk

AUCKLAND COUNCIL Rodney Section

COMPRISED IN CT	NA137A/506Ltd	57.4262ha
	NA105B/748	24.1140ha
	NA31C/965	21.6507ha
	NA757/204Ltd	37.0692ha
	NA2100/21Ltd	160.87ha
	NA127B/898	1.1705ha
TOTAL AREA		302.3006ha

LEVELS ARE IN TERMS OF LANDS AND SURVEY DATUM, AREAS AND MEASUREMENTS ARE SUBJECT TO SURVEY.

- Riparian Margin Planting
- Significant Ecological Area
- Proposed Building Area
- Watercourse
- Extent of Flood Plain

REVISION (DESCRIPTIONS)	NAME	DATE
R4 Bdys Lots 1 & 6 amended, Lot 8 added	LS	01.18
R3 Bdys Lots 1,2 & 6 amended	LS	12.17
R2 Flood Plain Added Lots 1 & 2	LS	11.17

	NAME	DATE
SURVEYED		
DESIGNED		
DRAWN	LS	09.17
CHECKED	MG	09.17
APPROVED	MG	09.17

CATO BOLAM CONSULTANTS

SURVEYORS PLANNERS ENGINEERS

CATO BOLAM CONSULTANTS LTD
19 Tamariki Avenue
PO Box 157
Orewa 0946
phone 09-427 0972
fax 09-426 7331
email catobolam@catobolam.co.nz

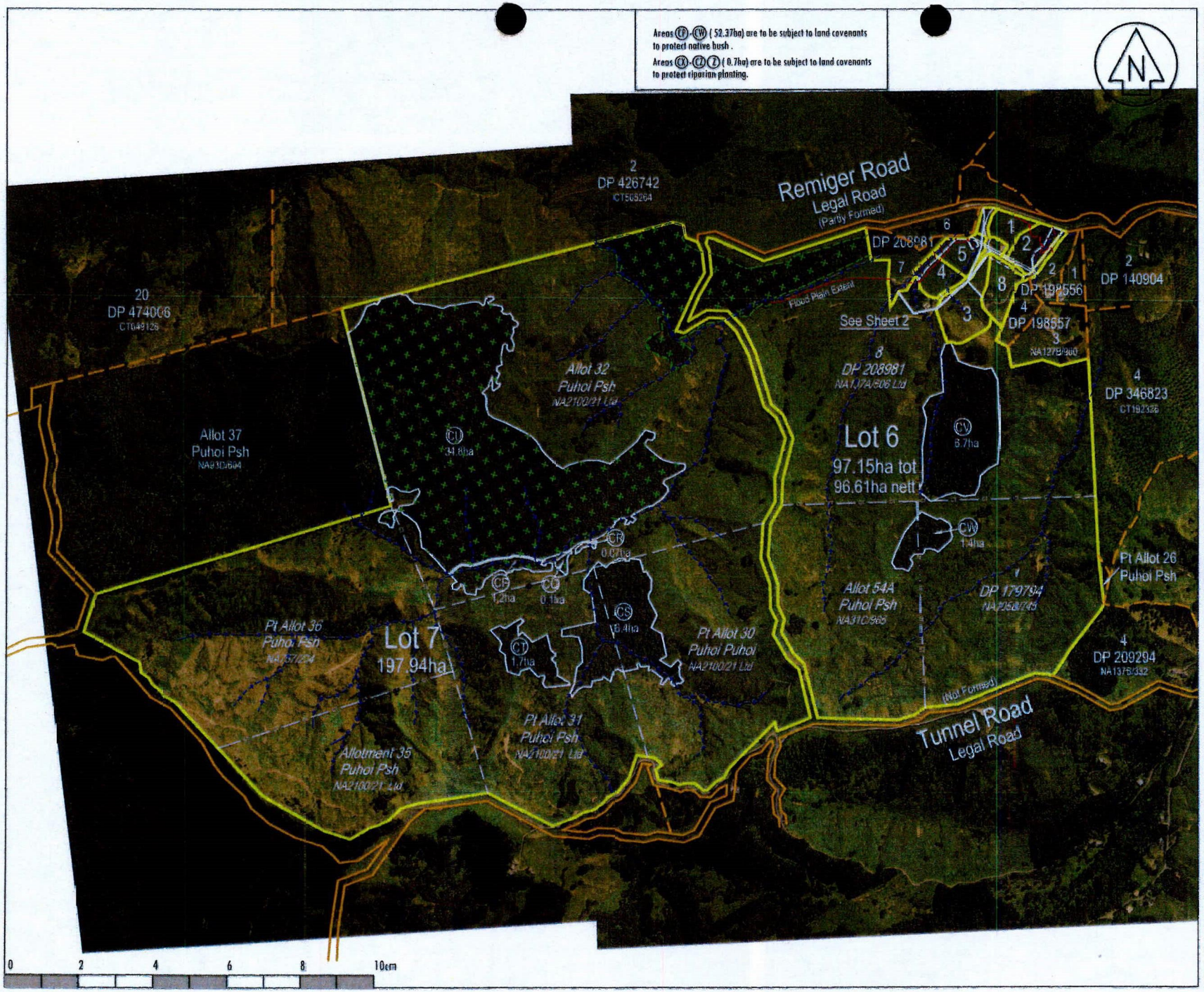
CLIENT

R.C. & D.E. NICKLIN
53B REMIGER ROAD
PUHOI

TITLE

LOTS 1 - 8 BEING
PROPOSED SUBDIVISION OF
LOT 8 DP 208981, LOT 5 DP 198556
LOT 1 DP 179794, ALLOTS 32,35,54A &
PT ALLOTS 30,31,36 PARISH OF PUHOI

ORIGINAL SCALE 1:10000	ORIGINAL SIZE A3	REVISION NO R4
DATE SEP 2017	CAD REFERENCE 40320-1200	SHEET NO 2 OF 2
DIRECTORY SYNERGY 40320\Drwgs\Schemes		JOB NO 1200 40320



This drawing and design remains the property of Cato Bolam Consultants ltd and may not be reproduced without the written permission of Cato Bolam Consultants ltd.

This plan and accompanying report(s) have been prepared for the purpose of obtaining a resource consent only and for no other purpose. Use of this plan and/or the information on it for any other purpose is at the user's risk

AUCKLAND COUNCIL Rodney Section

COMPRISED IN CT	NA137A/506Ltd	57.4262ha
	NA105B/748	24.1140ha
	NA31C/965	21.6507ha
	NA757/264Ltd	37.0692ha
	NA2100/21Ltd	160.87ha
	NA127B/898	1.1705ha
TOTAL AREA		302.3006ha

LEVELS ARE IN TERMS OF LANDS AND SURVEY DATUM, AREAS AND MEASUREMENTS ARE SUBJECT TO SURVEY.

- Riparian Margin Planting
- Significant Ecological Area
- Proposed Building Area
- Watercourse
- Extent of Flood Plain

REVISION (DESCRIPTIONS)	NAME	DATE
R4 Bdys Lots 1 & 6 amended, Lot 8 added	LS	01.18
R3 Bdys Lots 1,2 & 6 amended	LS	12.17
R2 Flood Plain Added Lots 1 & 2	LS	11.17

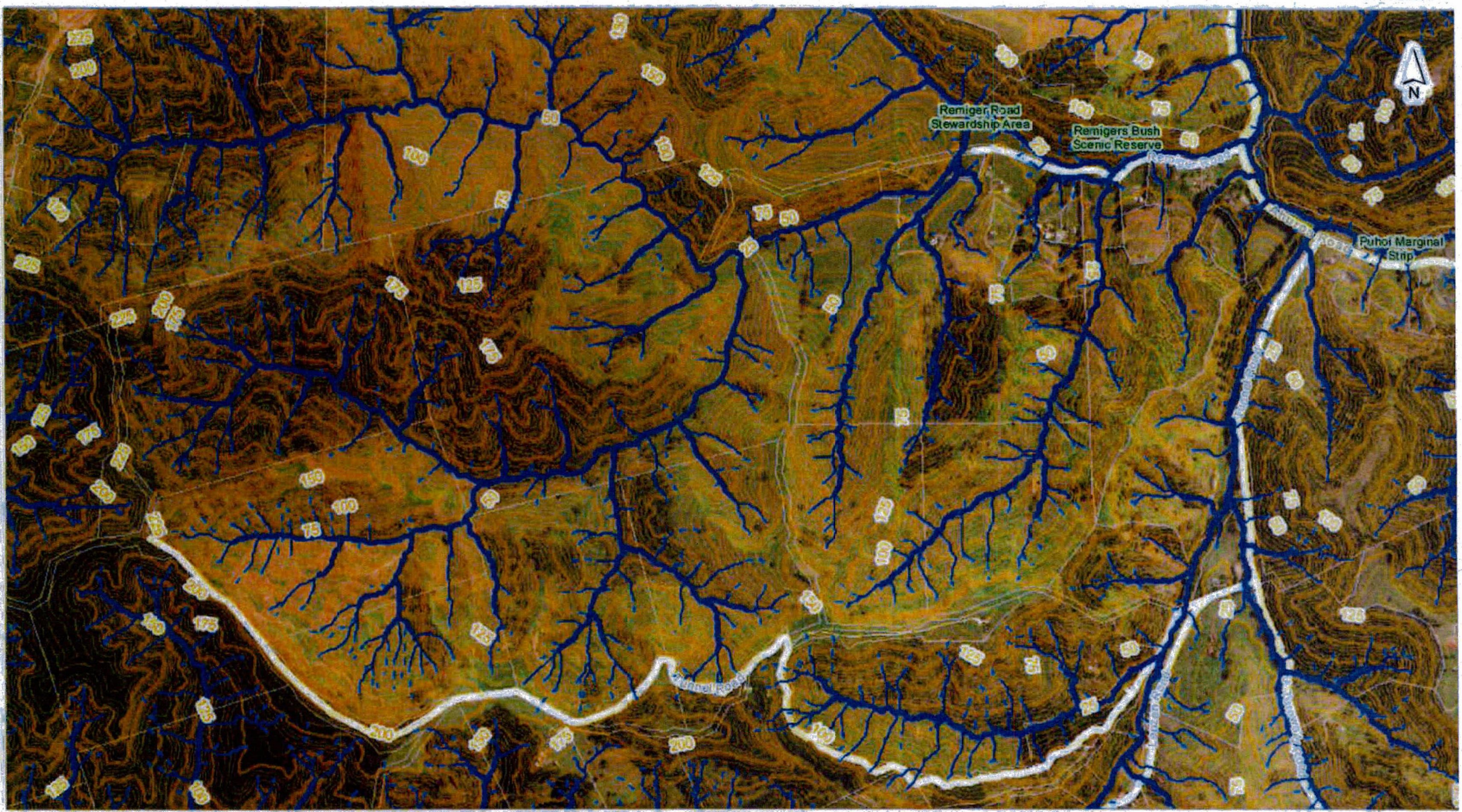
	NAME	DATE
SURVEYED		
DESIGNED		
DRAWN	LS	09.17
CHECKED	ME	09.17
APPROVED	ME	09.17

CATO BOLAM CONSULTANTS
SURVEYORS PLANNERS ENGINEERS
CATO BOLAM CONSULTANTS LTD
19 Tamariki Avenue
PO Box 157
Orewa 0946
phone 09-427 0072
fax 09-426 7331
email catobolam@catobolam.co.nz

CLIENT
R.C. & D.E. NICKLIN
53B REMIGER ROAD
PUHOI

TITLE
LOTS 1 - 3 BEING
PROPOSED SUBDIVISION OF
LOT 8 DP 208981, LOT 5 DP 198556
LOT 1 DP 179794, ALLOTS 32,35,54A &
PT ALLOTS 30,31,36 PARISH OF PUHOI

ORIGINAL SCALE 1:10000	ORIGINAL SIZE A3	REVISION NO R4
DATE SEP 2017	CAD REFERENCE 40320-12E0	SHEET NO 1 OF 2
DIRECTORY SYNERGY 40320\Drwgs\Schemes		JOB NO 1200 40320



DISCLAIMER
This map/plan is illustrative only and all information should be independently verified on site before taking any action. Copyright Auckland Council. Land Parcel Boundary information from LINZ (Crown Copyright Reserved). Whilst due care has been taken, Auckland Council gives no warranty as to the accuracy and plan completeness of any information on this map/plan and accepts no liability for any error, omission or use of the information. Height datum: Auckland 1946.

Tunnel Road, Puhoi CB Ref: 40320

0 100 200 300
Meters

Scale @ A4
= 1:15,000

Date Printed:
22/09/2017

Auckland Council
Te Kaunihera o Tāmaki Makaurau

Appendix B: TP108 Rainfall Calculations

Client	R C & D E Nicklin	Job No:	40320
Job Address	Tunnel Road		
	Puhoi	Date:	16-Oct-17

CATO BOLAM CONSULTANTS

PLANNERS | SURVEYORS | ENGINEERS | ENVIRONMENTAL

Stormwater Runoff Calculation Worksheet - Based on ARC TP 108, April 1999.

Undeveloped — Sheet 1 Pervious Fraction

1 Runoff Curve Number (CN) and Initial Abstraction (Ia)

Soil name and classification	Cover type, treatment, and hydrologic condition	Curve No. (CN)	Area (Ha)	Product of CN x Area
			Pervious	
Waitemata	Pasture	74	301.8460	22336.604
Group C	Protected Wetland	71	0.0000	0
	Protected bush	70	0.0000	0
		Total pervious	301.846	
		Totals	301.846	22336.604

CN (weighted) = total product / total area = **74**

Ia (weighted) = 5 x pervious area / total area = **5**

2 Time of Concentration

Channelisation factor - from table 4.2	C =	1
Catchment length (km)	L =	2.1950
Catchment slope (Fall in metres/length in metres)	S_c =	0.3000
Runoff factor CN / 200-CN	=	0.587302

Time of Concentration

$Tc = 0.14\ C\ L^{0.66}\ (CN/200\text{-}CN)^{-0.55}\ Sc^{-0.3}$ Tc = 0.452 hrs

Graphical Peak Flow Rate

1 Data

Catchment Area (km^2)	A =	3.01846	
Runoff curve number	CN =	74	
Initial abstraction	Ia =	5.0	
Time of concentration	T_c =	0.45	min 0.17hrs

2 Calculation of Storage

S = ((1000/CN)-10) 25.4 S = 89.2

		Storm 1	Storm 2	Storm 3	Storm 4	Storm 5	
3	Average recurrence interval (ARI) 24hr raintall depth, P_{24}(mm) (from TP 108)	1/3x2yr	34.5mm	2yr	10yr	100yr	(yrs)
4	(adjusted for climate change)	**36**	34.5	109	192	292	(mm)
5	Compute $c^* = P_{24}-(2 \times Ia)/(P_{24} - 2 \times Ia) +2 \times S$	0.13	0.12	0.36	0.50	0.61	
6	Specific peak flow rate q^* (from figure 5.1)	**0.029**	**0.028**	**0.067**	**0.085**	**0.095**	
7	Peak flow rate, $q_p = q^* \times A \times P_{24}$	3.2319	2.8922	22.2036	49.0512	83.5869	(m^3/s)
8	Runoff Depth, $Q_{24} = (P_{24} - Ia)^2/((P_{24}-Ia) + S)$	8	7	56	127	219	(mm)
9	Runoff volume, $V_{24} = 1000 \times Q_{24} A$	24,577	22,122	168,946	382,100	660,816	(m^3)

Client	R C & D E Nicklin	Job No:	40320
Job Address	Tunnel Road		
	Puhoi	Date:	16-Oct-17

CATO BOLAM CONSULTANTS

PLANNERS | SURVEYORS | ENGINEERS | ENVIRONMENTAL

Stormwater Runoff Calculation Worksheet - Based on ARC TP 108, April 1999.

Undeveloped — Sheet 2 Impervious Fraction

1 Runoff Curve Number (CN) and Initial Abstraction (Ia)

Soil name and classification	Cover type, treatment, and hydrologic condition	Curve No. (CN)	Area (Ha)	Product of CN x Area
			Pervious	
	Paved Areas	98	0.0375	3.675
	Roof	98	0.1370	13.426
	Gravel	89	0.2800	24.92
		Total pervious	0.4545	
				302.3
		Totals	0.4545	42.021

0.000

CN (weighted) = total product / total area = 92.45545

Ia (weighted) = 5 x pervious area / total area = 0

2 Time of Concentration

Channelisation factor - from table 4.2	C =	1
Catchment length (km)	L =	2.20
Catchment slope (Fall in metres/length in metres)	S_c =	0.30
Runoff factor CN / 200-CN	=	0.859694

Time of Concentration

$Tc = 0.14\ C\ L^{0.66}\ (CN/200\text{-}CN)^{-0.55}\ Sc^{-0.3}$ Tc 0.367 hrs

Graphical Peak Flow Rate

1 Data

Catchment Area (km^2)	A =	0.00455
Runoff curve number	CN =	92
Initial abstraction	Ia =	0.0
Time of concentration	T_c =	0.37 min 0.17hrs

2 Calculation of Storage

S = ((1000/CN)-10) 25.4 S = 20.7

		Storm 1	Storm 2	Storm 3	Storm 4	Storm 5	
3	Average recurrence interval (ARI)	1/3x2yr	34.5mm	2yr	10yr	100yr	(yrs)
	24hr rainfall depth, P24(mm) (from TP 108)						
4	(adjusted for climate change)	36.33333	34.5	109	192	292	(mm)
5	Compute $c^* = P_{24}-(2 \times Ia)/(P_{24} - 2 \times Ia) +2 \times S$	0.47	0.45	0.72	0.82	0.88	
6	Specific peak flow rate q* (from figure 5.1)	**0.093**	**0.091**	**0.123**	**0.131**	**0.134**	
7	Peak flow rate, $q_p = q^* \times A \times P_{24}$	0.0153	0.0143	0.0609	0.1139	0.1783	(m^3/s)
8	Runoff Depth, $Q_{24} = (P_{24} - Ia)^2/((P_{24}-Ia) + S)$	23	22	92	173	273	(mm)
9	Runoff volume, $V_{24} = 1000 \times Q_{24} A$	105	98	416	788	1239	(m^3)

Combined Flows and Run off Volumes

Storm	1/3x2yr	34.5mm	2yr	10yr	100yr	
24 hr rainfall depth (adjusted for climate change)	36	34.5	109	192	292	(mm)
Total Peak Flow rate (m3/s)	**3.247**	**2.906**	**22.264**	**49.165**	**83.765**	(m3/s)
Total Run off Volume (m3)	**24,683**	**22,220**	**169,362**	**382,888**	**662,055**	m3

Client	R C & D E Nicklin	Job No:	40320
Job Address	Tunnel Road		
	Puhoi	Date:	16-Oct-17

CATO BOLAM CONSULTANTS
PLANNERS | SURVEYORS | ENGINEERS | ENVIRONMENTAL

Stormwater Runoff Calculation Worksheet - Based on ARC TP 108, April 1999.

Developed — Sheet 1 Pervious Fraction

1 Runoff Curve Number (CN) and Initial Abstraction (Ia)

Soil name and classification	Cover type, treatment, and hydrologic condition	Curve No. (CN)	Area (Ha) Pervious	Product of CN x Area
Waitemata	Pasture	74	248.3260	18376.124
Group C	Protected Wetland	71	0.0000	0
	Protected bush	70	53.0700	3714.9
		Total pervious	301.396	
		Totals	301.396	22091.024

CN (weighted) = total product / total area = **73**

Ia (weighted) = 5 x pervious area / total area = **5**

2 Time of Concentration

Channelisation factor - from table 4.2 — C = 1

Catchment length (km) — L = 2.195

Catchment slope (Fall in metres/length in metres) — S_c = 0.30

Runoff factor CN / 200-CN = 0.578478

Time of Concentration
$Tc = 0.14\ C\ L^{0.66}\ (CN/200\text{-}CN)^{-0.55}\ Sc^{-0.3}$ — Tc = 0.456 hrs

Graphical Peak Flow Rate

1 Data

Catchment Area (km^2)	A =	3.01396
Runoff curve number	CN =	73
Initial abstraction	Ia =	5.0
Time of concentration	T_c =	0.46 min 0.17hrs

2 Calculation of Storage
S = ((1000/CN)-10) 25.4 — S = 92.5

		Storm 1	Storm 2	Storm 3	Storm 5	Storm 5	
3	Average recurrence interval (ARI) 24hr rainfall depth, P_{24}(mm) (from TP 108)	1/3x2yr	34.5mm	2yr	10yr	100yr	(yrs)
4	(adjusted for climate change)	11.5	34.5	109	192	292	(mm)
5	Compute $c^* = P_{24}-(2 \times Ia)/(P_{24} - 2 \times Ia) + 2 \times S$	0.01	0.12	0.35	0.50	0.60	
6	Specific peak flow rate q^* (from figure 5.1)	0.001	0.027	0.066	0.083	0.094	
7	Peak flow rate, $q_p = q^* \times A \times P_{24}$	0.0315	2.7920	21.7372	48.2718	82.5097	(m^3/s)
8	Runoff Depth, $Q_{24} = (P_{24} - Ia)^2/((P_{24}-Ia) + S)$	0	7	55	125	217	(mm)
9	Runoff volume, $V_{24} = 1000 \times Q_{24} A$	1286	21492	165863	377029	654097	(m^3)

Client	R C & D E Nicklin	Job No:	40320
Job Address	Tunnel Road		
	Puhoi	Date:	16-Oct-17

CATO BOLAM CONSULTANTS

PLANNERS | SURVEYORS | ENGINEERS | ENVIRONMENTAL

Stormwater Runoff Calculation Worksheet - Based on ARC TP 108, April 1999.

Developed — Sheet 2 Impervious Fraction

1 Runoff Curve Number (CN) and Initial Abstraction (Ia)

Soil name and classification	Cover type, treatment, and hydrologic condition	Curve No. (CN)	Area (Ha)	Product of CN x Area
			Pervious	
	Gravel	89	0.2800	24.92
	Roof	98	0.3620	35.476
	Paving	98	0.2625	25.725
		Total pervious	0.9045	
				302.3
		Totals	0.9045	86.121

0.000

CN (weighted) = $\frac{\text{total product}}{\text{total area}}$ = 95.21393

Ia (weighted) = $\frac{\text{5 x pervious area}}{\text{total area}}$ = 0

2 Time of Concentration

Channelisation factor - from table 4.2	C =	1	
Catchment length (km)	L =	2.20	
Catchment slope (Fall in metres/length in metres)	S_c =	0.30	
Runoff factor $\frac{CN}{200-CN}$	=	0.908651	
Time of Concentration $Tc = 0.14\ C\ L^{0.66}\ (CN/200-CN)^{-0.55}\ Sc^{-0.3}$	Tc	0.356	hrs

Graphical Peak Flow Rate

1 Data

Catchment Area (km^2)	A =	0.01	
Runoff curve number	CN =	95	
Initial abstraction	Ia =	0.00	
Time of concentration	T_c =	0.36	min 0.17hrs

2 Calculation of Storage

S = ((1000/CN)-10) 25.4 S = 12.8

		Storm 1	Storm 2	Storm 3	Storm 4	Storm 5	
3	Average recurrence interval (ARI)	1/3x2yr	34.5mm	2yr	10yr	100yr	(yrs)
	24hr rainfall depth, P24(mm) (from TP 108)						
4	(adjusted for climate change)	36	34.5	109	192	292	(mm)
5	Compute $c^* = P_{24}-(2 \times Ia)/(P_{24} - 2 \times Ia) +2 \times S$	0.59	0.57	0.81	0.88	0.92	
6	Specific peak flow rate q* (from figure 5.1)	0.112	0.111	0.131	0.136	0.139	
7	Peak flow rate, $q_p = q^* \times A \times P_{24}$	0.0367	0.0345	0.1292	0.2367	0.3665	(m^3/s)
8	Runoff Depth, $Q_{24} = (P_{24} - Ia)^2/((P_{24}-Ia) + S)$	27	25	98	180	280	(mm)
9	Runoff volume, $V_{24} = 1000 \times Q_{24} A$	243	228	883	1628	2530	(m^3)

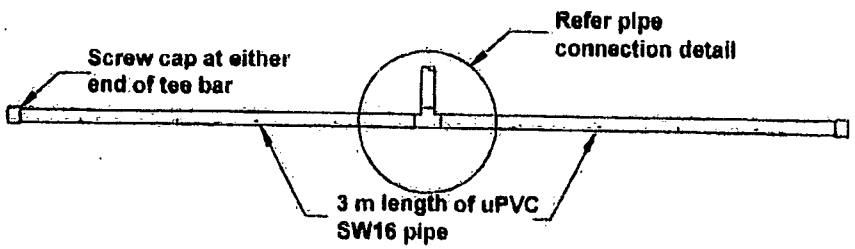
Combined Flows and Run off Volumes

Storm	1/3x2yr	34.5mm	2yr	10yr	100yr	
24 hr rainfall depth (adjusted for climate change)	36	34.5	109	192	292	(mm)
Total Peak Flow rate (m3/s)	0.068	2.827	21.866	48.508	82.876	(m3/s)
Total Run off Volume (m3)	1529	21,720	166,746	378,657	656,627	m3

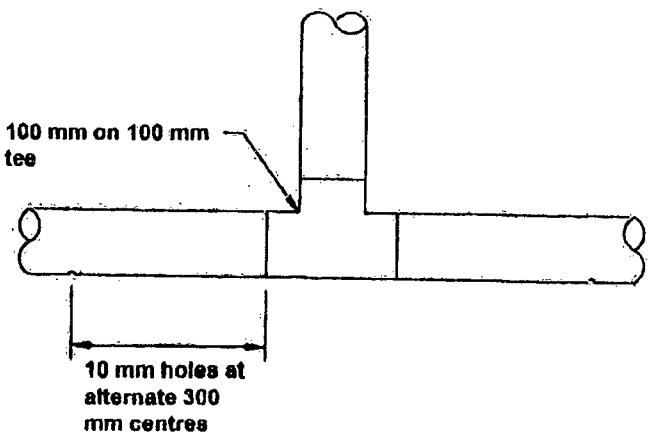
Pre and Post Development Summary

	ARI	34.5mm	Storm 1	Storm 2	Storm 3
Predev	Peak Flow (m^3/sec)	2.906	22.264	49.165	83.765
Postdev		2.827	21.866	48.508	82.876
	Change (m^3/sec)	0.080	0.398	0.657	0.889
Predev	Runoff Volume(m^3)	22,220	169,362	382,888	662,055
Postdev		21,720	166,746	378,657	656,627
	Change (m^3)	-500	-2,617	-4,231	-5,428

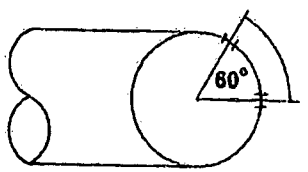
Appendix C: Level Spreader Device Diagrams



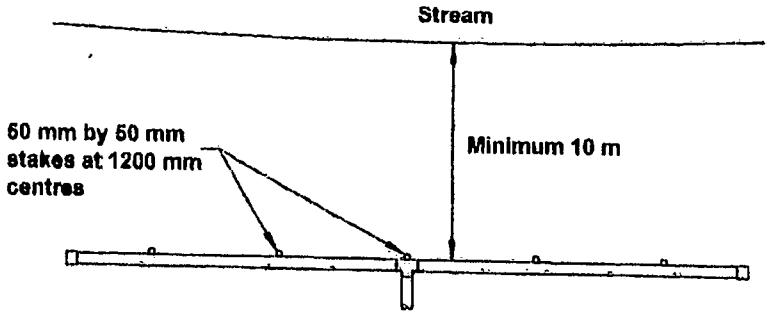
Plan



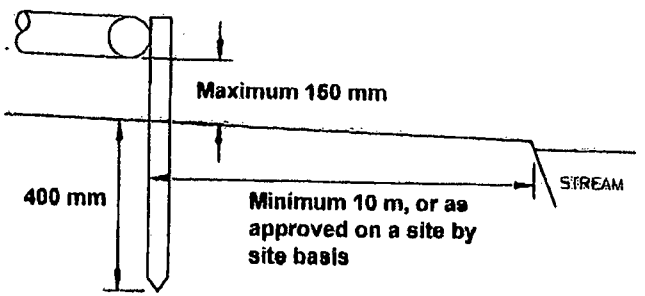
Pipe connection detail



Pipe outlet hole Arrangement detail



Layout Detail

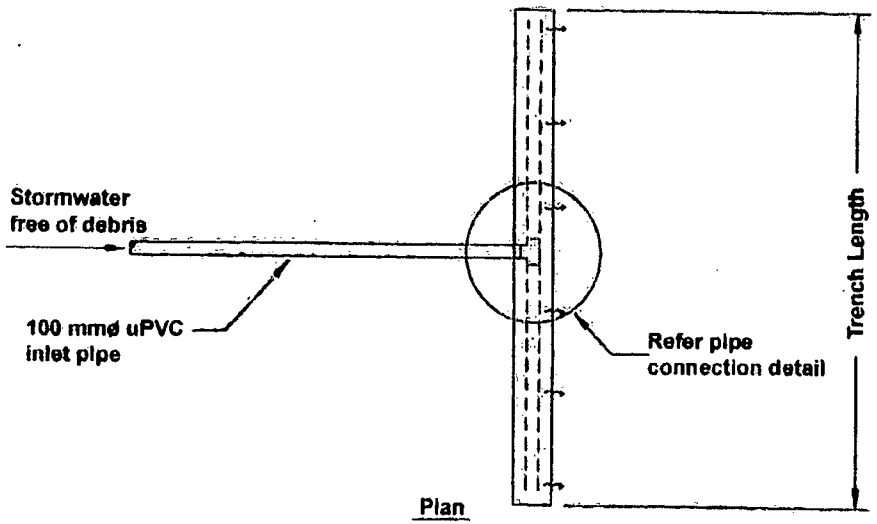


Stake Detail

Notes

1. Pin pipe and tee bar using 50x50 mm 114 treated stakes at 1200 mm centres. Level pipe and fix to stakes using down pipe clips.
2. Stakes to be driven 400 mm into ground. Tee bar to be no more than 150 mm above ground level at any point and to be constructed dead level across length of tee bar.

Figure C22
Above Ground Flow Dispersal



Keep dispersal trench 30 m clear of property boundary

Design Parameters

Effective catchment area drained (m^2)	Trench Length (m)
100	8
200	12
300	14
400	16
500	18
600	20

(Note: effective catchment area drained = impervious area + 0.72 x pervious area

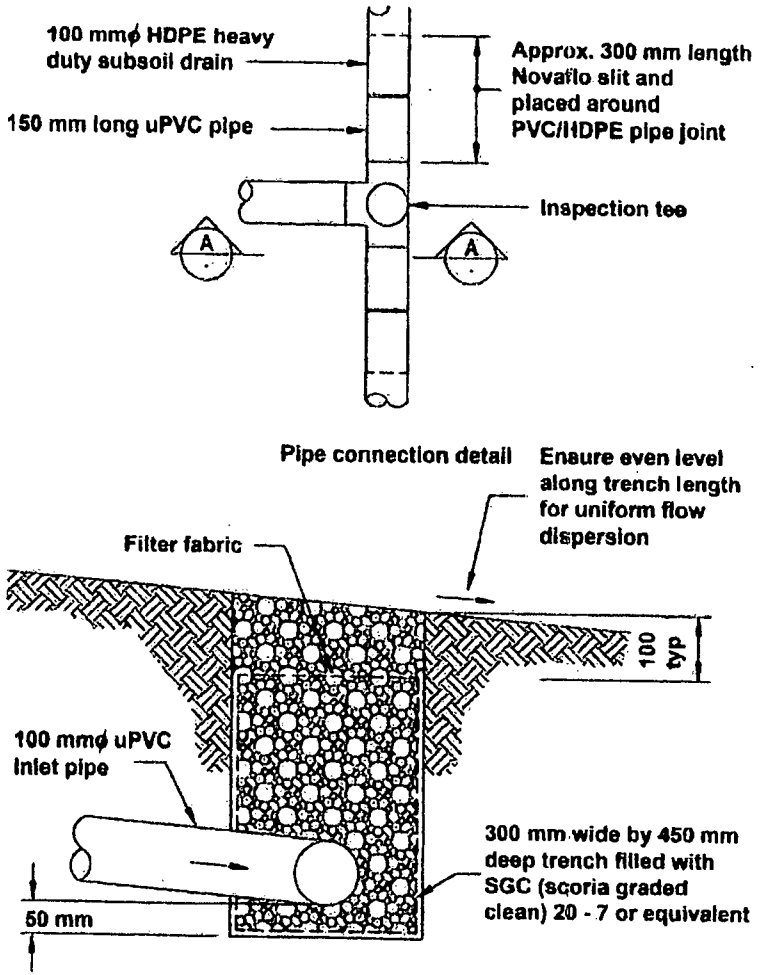


Figure C21
Conceptual Layout of Flow Dispersal Trench

Appendix D: Stormwater Device Maintenance Plan

Maintenance Plan – Stormwater Devices

1. Maintenance and Responsibilities

The Stormwater devices and the ongoing maintenance of the devices are the responsibility of the property owner.

2. Details of Proposed Maintenance Methods

Litter - following storms and at no less than monthly intervals, an inspection of the inlet and outlet structures of the stormwater tank, catchpits and dispersal device will be conducted, in addition gutters, spouting and all mesh screens or other devices are to be inspected and cleaned. Only those who have attended an approved OSH formal training course may enter or directly supervise when it is required to enter the tank. Any litter or debris should be removed and disposed of.

Sludge Buildup Inspection - the buildup of sludge at the bottom of the tank will be required to be inspected annually and removed if necessary. Professionals may be appointed by the lot owner if there is excessive sludge build up. Disposal of sludge must comply with all local and regional control programs; only suitable disposal sites should be used.

Working in confined spaces - only those who have attended an approved OSH formal training course or who are directly supervised by a person who has received formal training in working in a confined space may work in a confined space.

Structural Inspection - visual inspections of the tank, gutters, downpipes, catchpits, dispersal devices and structures should be carried out during litter inspections. If these inspections should identify that any repair or maintenance is required to the inlets and outlets, a qualified engineer or registered plumber should be appointed to advise of remedial works required. Repairs to any structural component of the devices should be made promptly.

3. Maintenance Frequency Schedule

Raintank, catchpits, gutters and dispersal device maintenance

The table below outlines the typical maintenance requirements and the frequency at which this maintenance should be carried out.

Item	Frequency	Completed by Whom
Litter check inlet and outlet, in particular the tank orifice and overflow and dispersal device outlets. Gutters, spouting, mesh screens, catchpits, swales and dispersal device flow outlets to avoid blockages.	Monthly and following storm Every six months	Lot owner completes all initial inspections
Sludge buildup inspections Structural Inspection	Annually Annually or as required by consent	Lot owner to appoint suitably qualified person if repairs or remedial work is required (refer to Hazard Control plans) C6 for hazards).